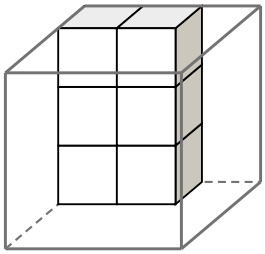
Math Kangaroo Practice

Grade 3-4 · 5 questions · standard · seed 61393

Part A: Each question is worth 3 points

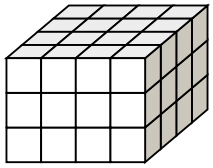
1. Leo puts cubes into a glass box, as in the picture. Every cube stands on the bottom or on another cube.



How many cubes are in the box?

A. 2 ***B.*** 4 ***C.*** 5 ***D.*** 6 ***E.*** 7

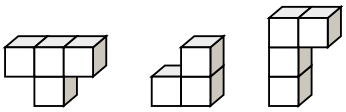
2. Leo built this brick (4 cubes long, 4 cubes deep and 3 cubes tall) from red and blue cubes of the same size. Every cube you can see from the outside (also from below and behind) is red. All the cubes inside are blue.

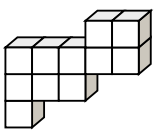


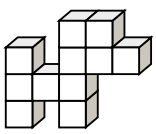
How many red cubes did Leo use?

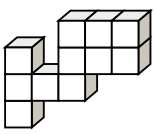
A. 30 ***B.*** 36 ***C.*** 40 ***D.*** 44 ***E.*** 48

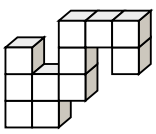
3. Kanga wants to build a flat figure out of these 3 parts made of cubes (the picture). He must use every part once, and he can turn the parts or flip them over. Which of these figures **can** he make?

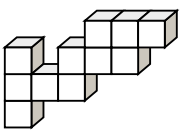


A. 

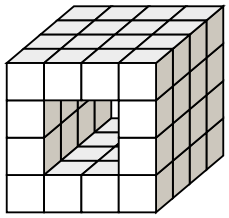
B. 

C. 

D. 

E. 

4. Sara built this tunnel out of identical cubes. The hole goes straight through from the front to the back.

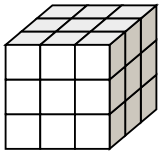


How many cubes did Sara use?

A. 64 ***B.*** 60 ***C.*** 48 ***D.*** 36 ***E.*** 12

Part B: Each question is worth 4 points

5. A cube made of 3 × 3 × 3 small cubes stands on the table and is painted grey on all sides you can reach: the top and the four sides, but **not** the bottom. Then it is taken apart.



How many small cubes have exactly 2 painted faces?

A. 16 ***B.*** 15 ***C.*** 13 ***D.*** 12 ***E.*** 8

Answer Key

1. **D**	2. **D**	3. **C**	4. **C**	5. **D**

Solutions

1. D (6) — Count each column from the bottom: 3 + 3 = 6. [set-03/by_level/Grade_03-04/2007 Q14 · count_inside]

2. D (44) — All cubes: 4 × 4 × 3 = 48. Blue inside: 2 × 2 × 1 = 4. Red: 48 – 4 = 44. [set-03/by_level/Grade_03-04/2003 Q9 · red_outside]

3. C — Every figure has 11 cubes, so count is not enough. Start with a cube that sticks out: only one part can fill it. Only one figure can be filled completely. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · can_make]

4. C (48) — The front has 4 × 4 – 2 × 2 = 12 cubes. The tunnel is 4 cubes long, so 12 × 4 = 48. [set-03/by_level/Grade_03-04/2002 Q20 · count_tunnel]

5. D (12) — Top edges without their corners: 4 × 1 = 4 (top and one side). Upright edges without the top corner: 4 × 2 = 8 (two sides; the bottom corners count here because their bottom is not painted). Total 12. [set-03/by_level/Grade_03-04/2007 Q17 · not_underneath]